BHUVANESHWARI M

Python Developer • FastAPI • Computer Vision
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PROFILE SUMMARY

A Python Developer with 2+ years of experience in developing backend applications, REST APIs, real-time dashboards, and computer vision solutions. Proficient in Python, FastAPI, PostgreSQL, SQLAlchemy, OpenCV, YOLOv8, NVIDIA DeepStream, and GStreamer. Experienced in database integration, real-time traffic monitoring, vehicle detection and classification, ANPR data analysis, MQTT-based communication, and video management systems. Hands-on experience with Linux server deployment, systemd services, and application integration. Seeking opportunities in Python Backend Development and Computer Vision Engineering.

TECHNICAL SKILLS

Backend: Python, FastAPI, Django, REST APIs, SQLAlchemy, Pydantic
Frontend: HTML, CSS, JavaScript, Jinja2, Bootstrap
Data & Messaging: PostgreSQL, MySQL, Redis, MQTT, AWS Kinesis Video Streams
Authentication & Security: RBAC, JWT, Session-Based Token Management
AI & Video: NVIDIA DeepStream, YOLOv8, PaddleOCR, OpenCV, GStreamer, FFmpeg, OpenVINO
Tools & Deployment: Git, Linux, Nginx, systemd, VS Code; AI assistants (Claude, ChatGPT, Cursor, GitHub Copilot)

WORK EXPERIENCE

Python Developer | COS Artificial Intelligence Pvt. Ltd., Madurai **Jun 2024 – Present**

- Build end-to-end project dashboards: frontend UI, FastAPI backend APIs, PostgreSQL integration, validation and reporting.
- Run continuous R&D to improve ATCC vehicle counting/classification and ANPR accuracy; benchmark DeepStream and OpenVINO pipelines on GPU vs CPU.
- Implemented RBAC and JWT-based authentication with Redis for session and token management.
- Validate detection accuracy and performance on live camera streams; deploy with Nginx and systemd.

Python Developer Intern | COS Artificial Intelligence Pvt. Ltd., Madurai **Dec 2022 – Apr 2023**

- Learned Django, RTSP streams, OpenCV, GStreamer and DeepStream; set up the ATCC dashboard and added report downloads.

Front-End Developer Intern | Vyukshathriya Technologix Pvt. Ltd., Madurai **Sep 2022 – Nov 2022**

- Built responsive web dashboards with HTML, CSS, JavaScript and Bootstrap, and deployed them on Netlify.

PROJECTS

ATMS 2021 & 2023 | Centralized Traffic Management Dashboard — In progress

Stack: *Python | FastAPI | PostgreSQL | Redis | MQTT | Leaflet | ApexCharts*

- Developed a centralized traffic management dashboard with RBAC, e-Challan tracking, event video retrieval, and MQTT integration for unified monitoring.

GATE-I | AI Vehicle Access Control — In progress

Stack: *Python, | FastAPI | PostgreSQL | DeepStream | YOLOv8 | PaddleOCR | RFID reader | boom barrier*

- Developing an AI-based vehicle access control system integrating ANPR, RFID validation, automated gate control, and audit logging.

Toll Sync | Toll Automation & Traffic Analytics

Stack: *Python | FastAPI | PostgreSQL | DeepStream| YOLOv8 | PaddleOCR | AWS Kinesis Video Streams*

- Integrated ATCC, ANPR, and TMS audit reporting to analyze vehicle counts, classification, FASTag mismatches, and toll traffic trends. .

Movable ATCC | Real-Time Traffic Counting & Classification

Stack: *Python | FastAPI | PostgreSQL | DeepStream | YOLOv8 | GStreamer*

- Developed a real-time vehicle counting and classification system with live camera monitoring, analytics, and offline data storage with cloud synchronization.

RVTL | Rajkot Vadinar Toll Plaza | Vehicle Monitoring | ATCC

Stack: *Python | FastAPI | PostgreSQL | YOLOv8 | GStreamer | SQLAlchemy | Pandas*

- Developed a toll-plaza reporting module with real-time analytics, downloadable reports, audit logs, and historical data restoration.

Video Converter Tool | Video Conversion & Compression Dashboard

Stack: *Python | FastAPI | FFmpeg | MySQL | Jinja2*

- Developed a video conversion and compression dashboard with configurable resolution, FPS, bitrate, CRF, and encoding presets.

Also Built:

- **MLFF Audit Dashboard:** Compared toll reports, FASTag transactions, and RFID/ANPR data to analyze vehicle classification accuracy and mismatches.
- **Sesha Plus Audi**t: Developed an AI-vs-RFID audit dashboard for vehicle data comparison.

EDUCATION

M.Sc. Information Technology | Fatima College Autonomous, Madurai | **CGPA 8.5** **2021 – 2023**
B.Sc. Information Technology | Fatima College Autonomous, Madurai | **CGPA 7.9** **2018 – 2021**